APPLICATION FOR ETHICS APPROVAL

Mae'r ddogfen hon hefyd ar gael yn Gymraeg / This document is also available in Welsh

When undertaking a research or innovation project, Cardiff Met staff and students are obliged to complete this form in order that the ethics implications of that project may be considered.

The document ***Ethics application guidance notes*** will help you complete this form and is available from the Ethics Governance Section of the Cardiff Met website. The School or Unit in which you are based may also have produced some guidance documents which you can access via your supervisor or School Ethics Coordinator.

PLEASE NOTE:
Participant recruitment or data collection MUST NOT commence until ethics approval has been obtained.

Completion of all sections of Part One and the Declaration is compulsory in all cases.
Forms which have been modified to delete or edit any of the questions will be returned to the applicant without being considered.

PART ONE

1A GENERAL INFORMATION	
Name of applicant:	Bothalage Amindu Ravihara
Supervisor (if student project):	Vindya Karunarathne
School / Unit:	ICBT Campus
Student number (if applicable):	CL/BSCSD/33/43
Programme enrolled on (if applicable):	BSc Hons Software Engineering
Project Title: If using a working title, it should convey what the project is about	Scouting Star - AI-Powered Sports Player Recruitment and Recommendation Platform
Expected start date of data collection:	2026/06/29
Approximate duration of data collection:	Getting the dataset from kegggle.com
Funding Body (if applicable):	
Other researcher(s) working on the project: If your collaborators are external to Cardiff Met, include details of the organisation they represent	N/A

1B SECURITY SENSITIVITY	
Will the project have security sensitive implications? NB: Any project which falls under this definition will be referred to the Prevent Co-ordinator by the Committee which is considering the application so it can be risk assessed in line with the Prevent Policy. Ethics approval will only be granted once the Prevent Co-ordinator is content	No
If yes, please explain what they are and the measures that are proposed to address them. (500 words max)	

1C NON-TECHNICAL SUMMARY

Please provide a non-technical summary of the project below (max 250 words)

Scouting Star is a web-based platform designed to modernize the way sports talent is discovered and recruited globally. The platform specifically targets football and cricket, two of the world's most widely played sports, and aims to connect talented athletes with coaches, scouts, clubs, and sports agents in a smarter and more accessible way.

Currently, many talented players remain undiscovered simply because they lack the right connections or do not have access to professional scouts. At the same time, coaches and recruiters spend enormous amounts of time and money searching for suitable players through traditional methods such as attending matches and relying on personal networks.

Scouting Star solves this problem by providing athletes with a dedicated digital space to create professional profiles, showcase their skills, upload match videos, and display their performance records. Recruiters can then search through these profiles or simply describe the type of player they are looking for in plain language, and the platform's intelligent system will automatically suggest the most suitable candidates.

The intelligence behind the platform uses a combination of language understanding and statistical analysis to match recruiter requirements with player profiles accurately. Player videos and images are stored securely in the cloud, ensuring fast and reliable access from anywhere in the world.

The platform benefits athletes by increasing their visibility regardless of where they live, and benefits recruiters by saving time and reducing costs. Ultimately, Scouting Star creates a fairer, faster, and more globally connected sports recruitment process.

1D DATA COLLECTION AND STORAGE

What types of data will be collected or created as part of the project? (max 250 words)

What types of data will be collected or created as part of the project?

The **Scouting Star** project will collect and generate several types of data during its development and operation. **Player profile data** will include personal details such as name, age, nationality, contact information, playing position, experience level, physical attributes, skill descriptions, and performance statistics relevant to the sport, such as goals, assists, batting averages, or wickets.

Media data will consist of player-uploaded images and highlight videos showcasing their performance. These files will be stored securely using Cloudinary cloud storage, while only their reference URLs will be maintained in the platform database.

Recruiter data will include account information for scouts, coaches, clubs, and agents, together with search queries, shortlisted players, and recruitment activities carried out within the platform.

To develop and evaluate the AI recommendation model, the project will use a **secondary dataset obtained from Kaggle**, containing publicly available sports performance data. This dataset will be used solely for AI training, validation, and testing. In addition, a small, labelled test dataset will be created to evaluate recommendation accuracy.

The system will also generate interaction data, including activity logs, recommendation history, and content moderation records. All collected data will be securely stored and used only for sports recruitment, AI model development, system evaluation, and research purposes.

How will you manage access to and security of the data? (max 500 words)

Data Access and Security Management

The Scouting Star platform will implement a comprehensive data access and security framework to ensure that all collected data is protected, appropriately restricted, and managed responsibly throughout the system's operation.

Role-Based Access Control

The platform operates three distinct user roles — Player, Scout/Coach, and Administrator — each with clearly defined access permissions. Players can only access and modify their own profile data and cannot view other users' private account information. Scouts and coaches can view approved player profiles and their own account data but cannot modify player information or access administrative functions. Administrators have the highest level of access, enabling them to manage accounts, moderate content, and monitor platform activity. This role-based structure ensures that no user can access data beyond what is necessary for their intended purpose.

Authentication and Password Security

All user accounts will be protected using JSON Web Token (JWT) authentication, which generates secure, time-limited session tokens upon login. Passwords will never be stored in plain text within the database. Instead, all passwords will be encrypted using the bcrypt hashing algorithm before storage, ensuring that even in the event of unauthorized database access, user passwords remain protected.

Database Security

The MySQL database will be configured to restrict direct external access, allowing connections only through the authenticated backend API. Database queries will be constructed using parameterized statements to prevent SQL injection attacks, which represent one of the most common vulnerabilities in web-based data systems. Sensitive personal data fields will be stored with appropriate access restrictions at the database level.

Media Storage Security

Player images and videos will be stored on Cloudinary's cloud storage platform rather than on the application server directly. Cloudinary provides built-in access controls, secure HTTPS delivery, and signed URL capabilities that restrict unauthorized access to private media files. The application database retains only URL references, reducing the risk associated with storing large volumes of sensitive media on the primary server.

Data Minimization and Purpose Limitation

The platform will collect only the data necessary for sports recruitment purposes. Players will not be required to submit sensitive personal data beyond what is relevant to their sporting profile. Recruiter search queries and recommendation history will be retained only for the duration necessary to support platform functionality and system improvement.

Account Approval and Content Moderation

All newly registered accounts require administrator approval before becoming active on the platform. This verification step reduces the risk of fraudulent accounts accessing the system and helps maintain the integrity of data used by the AI recommendation model. Administrators can suspend or permanently remove accounts found to violate platform policies.

Secure Development Practices

Throughout development, API endpoints will be tested using Postman to identify and resolve security vulnerabilities before deployment. The GitHub repository will be maintained with appropriate access controls to protect source code from unauthorized modification.

These combined measures ensure that data collected through the Scouting Star platform is accessed only by authorized users, stored securely, and managed in accordance with responsible data handling principles.

Will the project involve human samples and/or human cell lines?	No

Will the data collected be subject to the data retention protocols of either:

- Human Tissue Authority (HTA)
- The Health Research Authority - Health and Care Research Wales (HRA-HCRW)

Yes ☐

For any project which is subject to the data retention protocols of either the HTA or HRA-HCRW, you are required to develop a data storage plan to be submitted alongside your ethics application. The plan should

expand on the answers you have provided to the preceding two questions and should show how it will meet the expectations of the external body involved.
No ☒ Please confirm that the data collected will be stored in a manner which complies with Cardiff Met requirements via one of the following statements.

STATEMENT 1: FOR STUDENTS ON TAUGHT COURSES I confirm that any non-anonymised data related to research participants will only be stored on OneDrive, or by agreement with supervising staff, on Figshare, and that all data held elsewhere will be deleted, unless it is anonymised.	☒
STATEMENT 2: FOR STAFF APPLYING ON BEHALF OF STUDENTS ON TAUGHT COURSES I confirm that all students covered by this application are aware of their obligation to ensure that non-anonymised data related to research participants must only be stored on their Cardiff Met student OneDrive account and that all data held elsewhere must be deleted, unless it is anonymised.	☐
STATEMENT 3: FOR RESEARCH STUDENTS AND STAFF I confirm that any non-anonymised data related to research participants will be stored in a secure manner (using a platform such as OneDrive or FigShare) and that all data held elsewhere will be deleted unless it is anonymised.	☐

1E PROJECT CATEGORIES	
From the options provided, select the category which best describes your project, bearing in mind that "desk based" can encompass any non-practice based project which takes place primarily outside of a laboratory.	Desk based ☒ Laboratory based ☐ Practice based ☐
Do any of the following statements describe your project?: • Desk based involving only documents and ***NOT*** involving the collection of data from participants OR • Laboratory based ***NOT*** involving human participants, human samples, animals or animal derived material OR • Practice based (eg curatorial, practice audit) ***NOT*** involving human participants	Yes
Answering **YES** indicates that the project does not include any participants and you will not therefore be collecting participant data. If this is the case, complete the Declaration at the bottom of the form and forward this form to your School Ethics Committee (or equivalent). No further information regarding your project is required and you do not need to complete any more sections of this form. If you have answered **NO**, please proceed to the next question.	

1F NHS	
Does your project involve the NHS?	No

If no, proceed to 1G	

If yes, has the NHS deemed that your project requires an IRAS application?

Yes ☐

For a project which requires an IRAS application, you must submit the following alongside your ethics application.

- The study protocol document
- The completed IRAS form
- Any other documents relevant to the IRAS application

Failure to do this will result in your application being returned to you.

No ☒

Proceed to Part Two (below)

1G ETHICS APPROVAL OBTAINED ELSEWHERE

Has your project already received ethics approval from a body or organisation other than Cardiff Met?

Yes ☐

No further information regarding your project is required.

Complete the Declaration at the bottom of this form and submit it to your School Ethics Committee (or equivalent) alongside the following documents:

- Copy of the externally approved ethics application and any associated documents.
- Copy of the approval letter from the external organisation.

No ☒

Proceed to Part Two (below)

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PART TWO

RESEARCH DESIGN	
1 Will you be using an approved protocol in your project? (An approved protocol is one which has been approved by Cardiff Met to be used under supervision of designated members of staff. For details of protocols in use in your School or Unit, contact your Ethics Coordinator)	No
If yes, please state the name and code of the approved protocol to be used	

2 Describe the research design to be used in your project by giving details under the following four headings. If your project involves the use of an approved protocol, much less detail will be required but you should indicate which areas of the project are covered by the protocol

- The research method(s) to be used;
- Sample and sampling;
- The proposed participant group, including recruitment methods, activities to be undertaken, the expected time commitment and details of any proposed payments;
- Analytical techniques to be used.

(max 1000 words)

Research Design

The Research Methods to be Used

This project adopts mixed-methods research design combining applied software engineering, experimental machine learning development, and quantitative system evaluation. The research follows a Design Science Research (DSR) methodology, which is widely recognized as appropriate for projects that produce a functional information system as their primary output. DSR involves iterative cycles of designing, building, testing, and evaluating a system to address a real-world problem, making it well suited to the development of an AI-powered web platform.

The primary research method is system development and experimentation, following an Agile-inspired iterative development process encompassing requirement analysis, system design, frontend and backend development, AI model construction, integration, and performance testing. A secondary quantitative method is applied through experimental evaluation of the AI recommendation model using controlled offline evaluation procedures. A tertiary user-based evaluation method is conducted through structured usability testing with a small participant group.

Sample and Sampling

Two sampling strategies are applied in this project. For AI model evaluation, purposive sampling will be used to construct a labeled test dataset of 50 recruiter query and player profile pairs, deliberately designed to represent diverse realistic recruiter requirements across football and cricket. Sports datasets for training the AI model will be sourced from publicly available repositories such as Kaggle, selected based on completeness and relevance.

For user testing, convenience sampling will be adopted, targeting 10 to 15 participants with relevant sporting or recruitment backgrounds recruited through direct invitation to university peers and sports club contacts.

The Proposed Participant Group, Including Recruitment Methods, Activities, Time Commitment and Payments

The participant group will consist of 10 to 15 volunteers aged 18 and above with some familiarity with sports participation, coaching, or recruitment. No technical background is required. Participants will be recruited through personal communication via direct invitation, with participation made entirely voluntary.

Each participant will complete a structured usability testing session comprising a five minute platform orientation, completion of predefined tasks including manual player search, AI recommendation query submission, and player profile viewing, followed by completion of a System Usability Scale (SUS) questionnaire. The expected time commitment is 30 to 45 minutes per participant. Sessions will be conducted in person or remotely via video call.

No payments or financial incentives will be offered. All data will be anonymized before analysis and no personally identifiable information will be retained beyond research requirements. Informed consent will be obtained from all participants prior to testing, and ethical approval will be sought through the institutional research ethics process before any participant involvement begins.

Analytical Techniques to be Used

Exploratory Data Analysis (EDA) using Python Pandas and NumPy will be conducted on player datasets to examine distributions, identify missing values, and inform feature engineering. AI model performance will be measured using Precision@5, Recall@5, F1-Score, Mean Average Precision (MAP), and NDCG, with results compared against a TF-IDF baseline to quantify improvement. User testing results will be analyzed using SUS scoring benchmarks. Functional system testing will employ structured test cases covering all platform features including authentication, search, recommendations, media upload, and administrative functions.

3 Will the project involve deceptive or covert research?	Y No
If yes, give a rationale for the use of deceptive or covert research (max 500 words)	

PREVIOUS EXPERIENCE

1 What previous experience of research involving human participants, relevant to this project, do you have? (max 500 words)

Previous Experience of Research Involving Human Participants

Prior to undertaking this project, I gained relevant experience involving human participants through my previous academic research project titled Pet Point: An Intelligent Multi-Tasking System for Stray Animal Care and Management. This project was developed as part of my undergraduate academic programme and involved the design and development of a web-based platform aimed at improving the identification, reporting, rescue, and welfare management of stray dogs and animals within local communities.

As part of that research, I conducted structured user requirement gathering sessions with a group of participants including animal welfare volunteers, local community members, and veterinary support staff. These sessions involved semi-structured interviews and questionnaire-based surveys designed to understand user needs, existing challenges in stray animal reporting, and expectations for a digital management solution.

Following system development, I conducted usability evaluation sessions with participant groups to assess the platform's accessibility, ease of use, and practical effectiveness. Participants were asked to complete predefined tasks within the system and provide feedback through structured evaluation forms. Informed consent was obtained from all participants prior to involvement, and all collected data was handled confidentially and anonymized before analysis.

This experience provided me with a foundational understanding of ethical research practices involving human participants, including consent procedures, data privacy management, and structured usability evaluation methodologies. The skills developed through that project — particularly in participant recruitment, questionnaire design, usability testing, and qualitative feedback analysis — are directly applicable to the user evaluation phase of the Scouting Star research project.

2. **Student projects only**

What previous experience of research involving human participants, relevant to this project, does your supervisor have? (max 500 words)

My supervisor has substantial academic experience overseeing student research projects that involve human participants across a range of computing and information systems disciplines. This experience includes supervising undergraduate and postgraduate projects involving user requirement gathering, system usability evaluation, and structured participant testing using standardized instruments such as the System Usability Scale (SUS) and user satisfaction questionnaires. My supervisor is familiar with institutional ethical approval processes, informed

consent procedures, and responsible data handling practices required when conducting research with human participants, providing appropriate guidance and oversight throughout the participant involvement phases of this project.

POTENTIAL RISKS

Provide details of any risks you foresee related to the project and, for each risk, explain the methods you will use to reduce them. Ensure that you include details of measures that will be taken to ensure the safety of participants in the event that any identified risks arise.
Include details of risks to the participants, the researcher and the project as a whole.
NB: You must provide detail related to all risks identified in the previous question. Failure to do so could result in your application being returned to you.
(max 1000 words)

1. Risk of Participant Discomfort or Stress During Usability Testing
Participants involved in usability testing sessions may experience mild stress or discomfort if they encounter difficulties completing tasks on the platform or feel their performance is being evaluated. To reduce this risk, participants will be clearly informed before testing begins that the evaluation is assessing the system and not the individual. A brief orientation session will be provided to familiarise participants with the platform before tasks commence. Participants will be reminded at all times that they are free to withdraw from the session without consequence and without providing a reason. The researcher will maintain a calm and supportive environment throughout all testing sessions.

2. Risk to Participant Data Privacy and Confidentiality
Participants may be concerned about the confidentiality of personal information and feedback provided during usability testing sessions. To mitigate this risk, all participant data will be fully anonymized before analysis, meaning no personally identifiable information will be retained or reported in any form. Participants will be assigned anonymous identifier codes for data recording purposes. Informed consent forms will clearly explain how data will be collected, stored, used, and disposed of upon project completion. All collected data will be stored securely on password-protected devices accessible only to the researcher and supervising academic staff.

3. Risk of Unauthorized Access to Platform User Data
The Scouting Star platform collects personal information, performance statistics, and media files from registered users. There is a risk that this data could be accessed by unauthorized parties if adequate security measures are not implemented. To reduce this risk the following measures will be applied:

- All user passwords will be encrypted using the bcrypt hashing algorithm before storage, ensuring plain text passwords are never retained in the database.
- JSON Web Token (JWT) authentication will be used to manage secure, time-limited user sessions and prevent unauthorized account access.
- Database queries will be constructed using parameterized statements to prevent SQL injection attacks.
- Player media files will be stored on Cloudinary cloud storage rather than on the application server, reducing exposure of sensitive media to server-level security breaches.
- Direct external access to the MySQL database will be restricted, with all data interactions routed exclusively through the authenticated backend API.

4. Risk of Fraudulent or Harmful User-Generated Content
As the platform allows users to upload profile information, statistics, images, and videos, there is a risk that some users may submit false, exaggerated, or inappropriate content. This could harm the reliability of the AI recommendation system and damage trust among recruiters using the platform. To manage this risk, all newly registered player accounts will require administrator approval before becoming publicly visible. A community reporting mechanism will allow scouts and coaches to flag suspicious profiles for review. Administrators will have

the authority to remove inappropriate content or suspend accounts found to violate platform policies. These measures will protect both the integrity of the platform and the experience of all users.

5. Risk to Researcher Wellbeing During Remote Testing Sessions
Where usability testing sessions are conducted remotely via video call, there is a minor risk related to researcher wellbeing and professional boundaries during unsupervised online interactions with participants. To mitigate this risk, all remote sessions will be conducted during standard working hours using established and secure video conferencing platforms. The researcher will inform their academic supervisor of all scheduled remote testing sessions. Session recordings will only be made with the explicit consent of the participant and will be deleted following data extraction.

6. Risk of Project Delays Affecting Data Collection and Evaluation
Technical challenges during system development, such as integration failures between the frontend, backend, and AI server, could delay the completion of the platform and reduce the time available for participant testing and AI model evaluation. To reduce this risk, the project follows a structured 12-week implementation plan with clearly defined milestones and buffer periods built into the later development phases. Regular progress reviews with the academic supervisor will ensure that delays are identified early and addressed promptly. If significant delays occur, the scope of usability testing will be adjusted accordingly while maintaining the integrity of the AI model evaluation phase.

When submitting your application, you **MUST** attach a copy of all information sheets, the consent/assent form(s) and the withdrawal of consent form.
An exemplar information sheet, exemplar participant consent form and exemplar participant withdrawal form are available via the research section of the Cardiff Met website (see section on Ethics Governance). These are based on good practice and will be useful in the majority of cases. However, it is recognised that in some cases a project will be subject to requirements from an external body. Use of these exemplars is therefore not obligatory.

APPLICATION FOR ETHICS APPROVAL

DECLARATION

I confirm that this project conforms with the Cardiff Met Research Integrity & Governance Framework **I confirm that I will abide by the Cardiff Met requirements regarding confidentiality and anonymity when conducting this project.** **STUDENTS: I confirm that I will not disclose any information about this project without the prior approval of my supervisor.**	
Signature of the applicant:	Date:
FOR STUDENT PROJECTS ONLY	
Name of supervisor:	Date:
Signature of supervisor:	

Research Ethics Committee use only	
Decision reached: Click here to enter text.	
Project reference number: Click here to enter text.	
Name: Click here to enter text.	Date: Click here to enter a date.
Details of any conditions upon which approval is dependant: Click here to enter text.	